

LENA LIEBICH

HARVARD BUSINESS SCHOOL AI INSTITUTE, DIGITAL RESKILLING LAB

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Academic Appointments

Postdoctoral Fellow, Digital Reskilling Lab Harvard Business School AI Institute, Boston, MA	2026 – present
Research Affiliate Leibniz Institute for Financial Research SAFE, Frankfurt am Main, Germany	2026 – present

Education

PhD in Economics, <i>summa cum laude</i> Graduate School of Economics, Finance and Management (GSEFM), Goethe University Frankfurt, Germany Dissertation: <i>Essays on Human-AI Interaction and Household Finance</i> Committee: Christine Laudenbach, Michael Kosfeld, Guido Friebe, Florian Hett	2020 – 2026
Visiting Scholar Centre for Brain, Mind and Markets, University of Melbourne, Australia Two-month research stay at the invitation of Carsten Murawski.	2025 – 2026
Visiting Scholar Department of Decisions, Operations and Technology, CUHK Business School, Hong Kong SAR Two-month research stay at the invitation of Jenny Jin.	2025
MSc in Quantitative Economics Graduate School of Economics, Finance and Management (GSEFM), Goethe University Frankfurt, Germany Focus: applied econometrics, household finance, experimental finance.	2020 – 2025
BSc in Economics University of Bonn, Germany Focus: experimental economics, game theory, contract theory, applied microeconomics.	2016 – 2020

Research Interests

Human-AI interaction, experimental and behavioral economics, organizational economics

Publications

[1] **Knowing (Not) to Know: Explainable Artificial Intelligence and Human Metacognition**
von Zahn, M., Liebich, L., Jussupow, E., Hinz, O., & Bauer, K. (2025)
Forthcoming in Information Systems Research, published online in Articles in Advance.

Abstract: The use of explainable artificial intelligence (XAI) to render black-box artificial intelligence (AI) more interpretable to humans is gaining practical relevance. Prior research has shown that XAI can influence how humans “think”. Yet, little is known about whether XAI also affects how people “think about their thinking” (i.e., their metacognitive processes). We address this gap by investigating whether XAI affects metacognitive processes that guide human confidence judgments about their ability to perform a task and thereby, their decision whether to delegate the task to AI. We conducted two incentivized experiments in which domain experts repeatedly performed prediction tasks, with the option to delegate each task to an AI. We exogenously varied whether participants initially received explanations about AI’s overall prediction logic. We find that AI explanations improve how well humans understand their own ability to perform the task (metacognitive accuracy). This improvement causally increases both the frequency and effectiveness of human-to-AI delegation. Additional analyses show that these effects primarily occur when explanations reveal to humans that AI’s prediction logic diverges from their own, leading to a systematic reduction of overconfidence. Our findings highlight metacognitive processes as a central, previously overlooked channel through which AI explanations can influence human-AI collaboration. We discuss practical implications of our results for organizations implementing XAI to comply with regulatory transparency requirements as, for example, outlined in the European Union Artificial Intelligence Act.

PAPER

DOI

[2] Can GPT Mimic Human Preferences? An Empirical and Structural Investigation

Bauer, K., Liebich, L., & Kosfeld, M. (2025)

Proceedings of the European Conference on Information Systems (ECIS 2025)

Abstract: We examine the depth of large language models’ (LLMs) potential to serve as tools for simulating human decision-making processes. Using GPT-4 Turbo in a sequential prisoner’s dilemma, we find significant discrepancies between LLM-generated and human behaviors: GPT consistently exhibits a higher propensity to cooperate and forms overly optimistic expectations of human cooperation, leading to outcomes that diverge from observed human decision-making patterns. Yet, GPT’s decisions do not appear random; they align with a formal model of human preferences for fairness and efficiency that explains the behavior of most human participants in similar settings. We find a striking contrast: while GPT may fall short as accurate surrogate for human decisions at the surface level, the model nonetheless exhibits a capacity to reflect the underlying preferences that drive human behavior in strategic contexts. Our findings have implications for LLMs’ potential to serve as a managerial or research tool in simulating human behaviors.

PAPER

Work in Progress

[3] Responsive Advisors, Indifferent Clients: Evidence from a Field Experiment on the Role of Explainable AI in Financial Advice (Working Title)

Liebich, L., Bauer, K., Yang, C., Li, X. & Hinz, O. (2026)

[4] Personalized Communication and Repayment (Working Title)

Hett, F., Gill, A., Laudenbach, C., Liebich, L. & Siegel, S. (2026)

Policy Work (Pre-PhD)

[5] Unconventionally Green: Monetary Policy between Engagement and Conflicting Goals

Liebich, L., Nöh, L., Rutkowski, F., & Schwarz, M. (2023)

Review of Economics, 74(1), 53–77.

Abstract: We study the potential role of the ECB in supporting the transition to a low-carbon economy by decarbonizing the corporate sector purchase programme (CSPP). We demonstrate that the carbon intensity of CSPP purchases is determined by three main factors: First, by the CSPP-eligibility criteria, as these tend to exclude bonds from low-emission sectors. Second, by the underlying structure of the bond market, as this tends to be skewed towards carbon-intensive sectors. Third, among the eligible bonds, the ECB tends to select those from relatively emission-intensive sectors. To decarbonize the CSPP, the ECB can theoretically act along these three lines: (i) adjust the CSPP-eligibility criteria to expand the range of eligible low-carbon assets, (ii), revise the principle of market neutrality to tilt the CSPP

portfolio towards low-carbon companies, or, (iii), purchase so far neglected low-carbon bonds within the current eligibility and market neutrality framework. As all approaches have either very limited effects or are associated with significant theoretical and practical concerns, we conclude that their contribution to the success of active green monetary policy is not guaranteed, while at the same time risks arise for a monetary policy that targets price stability.

PAPER

DOI

[6] Current Developments in Green Finance

Liebich, L., Nöh, L., Rutkowski, F., & Schwarz, M. (2020)

German Council of Economic Experts, Working Paper 05/2020.

Abstract: The transformation of economies towards significantly reduced CO2 consumption raises high investment and capital requirements. Financial and capital markets can help to mobilize the necessary funds for global investment needs and to steer capital towards sustainable investments. Moreover, potential disruptive impacts of climate change on the financial system have started to become more apparent recently and require central banks, regulators and supervisors to take a conscious look at the risks and opportunities of climate change for financial intermediaries and markets. This article offers a comprehensive discussion on how green finance has been evolving thus far and explores the opportunities and key developments ahead with particular emphasis on four selected highly topical issues: 1) the introduction of German green government bonds, 2) obstacles to the correct pricing of climate-related risks, 3) the EU taxonomy that has recently been put forward to develop a uniform classification of sustainable economic activities as well as 4) the role of central banks in fostering the transition to a low-carbon economy.

PAPER

Teaching

Teaching Assistant, *Machine Learning for Management Decisions*

2025

MBA course with Kevin Bauer, Gutenberg School of Business, JGU Mainz

Teaching Assistant, *Financial Markets and Institutions*

2019

BSc Economics course with Isabel Schnabel, University of Bonn

Awards & Scholarships

Award for Excellence in Teaching

2026

Joint with Kevin Bauer, for the MBA course *Machine Learning for Management Decisions*, Gutenberg School of Business, JGU Mainz

Best Discussion Award

2025

9th SAFE Household Finance Workshop, sponsored by Behavioural Finance e.V. (€1,000)

First-Year PhD Scholarship

2020 – 2021

Leibniz Institute for Financial Research SAFE

Conferences, Seminars and Invited Talks

- 4th Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2026
- Digital Economy Workshop (presentation)
- SAFE Brown Bag Seminar (presentation)
- SAFE Junior Scholars Workshop: The New World of Corporate Boards (presentation) 2025
- 3rd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation)
- University of Oxford, Faculty of Law (presentation)
- 9th SAFE Household Finance Workshop (discussion)
- CUHK Business School (presentation)
- 38th Australasian Finance and Banking Conference, UNSW Business School (presentation)

• 2nd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) • SAFE Brown Bag Seminar (presentation) • 9th Spring School in Behavioral Economics, UCSD Rady School of Management (poster) • Machine+Behavior Conference, MPI / Toulouse School of Economics / University of British Columbia (poster) • 8th SAFE Household Finance Workshop (discussion) • EconTribute Workshop on Human-AI Collaboration, University of Cologne (lightning talk)	2024
• 1st Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) • PhD course <i>Subjective Beliefs, Attention and Economic Behavior</i>, University of Copenhagen (poster) • 1st Bonn-Frankfurt-Mannheim PhD Conference (presentation) • Workshop on Interactions of Humans and Algorithms, TU Berlin (poster) • SAFE Brown Bag Seminar (presentation) • Society for Experimental Finance Summer School, University of Sofia • 7th SAFE Household Finance Workshop (presentation) • CORA Conference on Robots and Automation, Goethe University (presentation) • ZEW / University of Mannheim Experimental Seminar (presentation)	2023
6th SAFE Household Finance Workshop (discussion)	2022

Academic Service

Supervision of Master's and Bachelor's theses	2022 – 2026
Department of Household Finance, Leibniz Institute for Financial Research SAFE	
Representative of PhD Researchers	2022 – 2024
Leibniz Institute for Financial Research SAFE	
Member of Student Representatives	2020 – 2024
Graduate School of Economics, Finance and Management (GSEFM)	
Organizing Committee, 1st Bonn-Frankfurt-Mannheim PhD Conference	2023
Keynote by Ulrike Malmendier, Goethe University Frankfurt, Germany.	
Referee	2025
European Conference on Information Systems (ECIS)	

Professional Experience

Junior Researcher / Doctoral Student	2022 – 2025
Department of Household Finance, Leibniz Institute for Financial Research SAFE, Frankfurt am Main, Germany	
Academic Researcher	2022 – 2023
European Central Bank, DG Macroeconomic Policy and Financial Stability, Frankfurt am Main, Germany	
Research on market-based indicators of climate systemic risk.	
Trainee	2022
European Central Bank, DG Macroeconomic Policy and Financial Stability, Frankfurt am Main, Germany	
Empirical analysis of euro-area banks' exposure to climate-related systemic risk following the CRISK methodology of Jung, Berner and Engle.	

Student Research Assistant	2021 – 2022
Leibniz Institute for Financial Research SAFE, Frankfurt am Main, Germany Data analysis and literature search supporting the faculty's research projects.	
Research Intern	2020
Scientific Staff of the German Council of Economic Experts (GCEE), Wiesbaden, Germany Empirical analysis of the carbon intensity of the ECB's bond purchases under the Corporate Sector Purchase Programme, contributing to GCEE working papers 05/2020 and 05/2021.	
Consulting Assistant	2019 – 2020
CPC Unternehmensmanagement AG, Bonn & Frankfurt am Main, Germany Conceptual framework for blended-learning experiences in HR applications.	
Consulting Intern	2018
CPC Unternehmensmanagement AG, Frankfurt am Main, Germany Organisational change management concept for a large logistics firm; data analysis and visualisation.	
Student Assistant, Editorial Office <i>Documentaries</i>	2017 – 2018
phoenix / ZDF, Bonn, Germany	
German Language Tutor (gap year)	2015 – 2016
German School Abroad Bilbao, Spain	

Affiliations

- Harvard Business School AI Institute
- Leibniz Institute for Financial Research SAFE (Research Affiliate)
- Behavioural Finance e.V. (Member)
- Mentoring Hessen, ProCareer.Doc Program (2024 cohort)

Personal

Stata, Python, LaTeX, Git, oTree, MS Office	Software
German (native), English (fluent), Spanish (fluent)	Languages
German	Citizenship

References

Available on request.